Full simulation tables for the primary functionals

The main text summarizes the principal simulation findings and shows coverage across conditions. This appendix provides the complete numerical results for the three primary functionals—ab, ψ_{speed}, and ω_{speed}—at $N = 100$, 200, and 500, including coverage, left- and right-tail noncoverage, and median interval width.

Table 1: Simulation results for the indirect effect ab at $N = 100$: coverage of nominal 95% intervals, tail balance, and median width, by condition and method ($S = 1{,}000$ data sets per cell, $B = 1{,}000$ shared bootstrap weight vectors).

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	89.7	0.2	10.1	0.216
Wald (HW)	90.0	0.1	9.9	0.229
Monte Carlo (HW)	94.0	0.3	5.7	0.246
IJ1 percentile	94.1	0.3	5.6	0.243
HOIJ-2 percentile	94.8	0.9	4.3	0.249
Bootstrap percentile	95.3	0.4	4.3	0.258
Normal data, misspecified model				
Wald (Expected)	90.0	0.4	9.6	0.226
Wald (HW)	91.2	0.6	8.2	0.239
Monte Carlo (HW)	93.7	0.9	5.4	0.257
IJ1 percentile	94.4	0.7	4.9	0.256
HOIJ-2 percentile	94.6	1.3	4.1	0.264
Bootstrap percentile	95.3	0.8	3.9	0.272
Non-normal data, correct model				
Wald (Expected)	88.8	0.3	10.9	0.210
Wald (HW)	90.2	0.3	9.5	0.219
Monte Carlo (HW)	93.5	0.5	6.0	0.237
IJ1 percentile	93.2	0.5	6.3	0.233
HOIJ-2 percentile	94.1	0.8	5.1	0.238
Bootstrap percentile	95.1	0.5	4.4	0.245
Non-normal data, misspecified model				
Wald (Expected)	89.3	0.7	10.0	0.219
Wald (HW)	90.3	0.7	9.0	0.230
Monte Carlo (HW)	93.0	0.8	6.2	0.245
IJ1 percentile	93.2	0.7	6.1	0.247
HOIJ-2 percentile	94.5	1.1	4.4	0.249
Bootstrap percentile	95.4	0.9	3.7	0.257

Note. Coverage, miss-left, and miss-right entries are percentages, based on $S = 1{,}000$ simulated data sets per cell. For a well-calibrated equal-tailed 95% interval, the reference noncoverage rates are 2.5% in each tail. Differences between HOIJ-2 and the bootstrap describe differences between the procedures; differences from 2.5% describe inferential tail imbalance.

Table 2: Simulation results for the indirect effect ab at $N = 200$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	92.0	0.4	7.6	0.153
Wald (HW)	92.8	0.4	6.8	0.156
Monte Carlo (HW)	95.0	1.0	4.0	0.161
IJ1 percentile	95.5	0.6	3.9	0.159
HOIJ-2 percentile	95.0	1.4	3.6	0.160
Bootstrap percentile	95.6	1.0	3.4	0.165
Normal data, misspecified model				
Wald (Expected)	92.9	0.5	6.6	0.158
Wald (HW)	94.0	0.6	5.4	0.162
Monte Carlo (HW)	95.4	0.9	3.7	0.166
IJ1 percentile	95.7	0.7	3.6	0.166
HOIJ-2 percentile	95.8	1.4	2.8	0.167
Bootstrap percentile	96.1	1.3	2.6	0.171
Non-normal data, correct model				
Wald (Expected)	91.3	0.9	7.8	0.154
Wald (HW)	91.7	0.3	8.0	0.159
Monte Carlo (HW)	93.4	0.7	5.9	0.163
IJ1 percentile	92.7	1.1	6.2	0.162
HOIJ-2 percentile	92.6	1.6	5.8	0.164
Bootstrap percentile	93.3	1.6	5.1	0.169
Non-normal data, misspecified model				
Wald (Expected)	90.8	0.2	9.0	0.157
Wald (HW)	91.6	0.1	8.3	0.165
Monte Carlo (HW)	93.7	0.6	5.7	0.168
IJ1 percentile	94.0	0.4	5.6	0.168
HOIJ-2 percentile	94.8	0.8	4.4	0.167
Bootstrap percentile	95.4	0.7	3.9	0.173

Note. See Table 1.

Table 3: Simulation results for the indirect effect ab at $N = 500$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	93.1	0.9	6.0	0.096
Wald (HW)	93.3	0.9	5.8	0.098
Monte Carlo (HW)	94.2	1.6	4.2	0.098
IJ1 percentile	94.7	1.4	3.9	0.098
HOIJ-2 percentile	94.7	1.7	3.6	0.098
Bootstrap percentile	95.0	1.6	3.4	0.099
Normal data, misspecified model				
Wald (Expected)	94.2	1.0	4.8	0.099
Wald (HW)	94.5	1.0	4.5	0.099
Monte Carlo (HW)	95.5	1.1	3.4	0.099
IJ1 percentile	95.6	1.1	3.3	0.099
HOIJ-2 percentile	95.6	1.4	3.0	0.099
Bootstrap percentile	96.1	1.4	2.5	0.101
Non-normal data, correct model				
Wald (Expected)	92.2	1.5	6.3	0.095
Wald (HW)	93.0	0.8	6.2	0.099
Monte Carlo (HW)	94.1	1.2	4.7	0.099
IJ1 percentile	93.9	1.7	4.4	0.100
HOIJ-2 percentile	93.5	2.1	4.4	0.099
Bootstrap percentile	94.0	1.9	4.1	0.100
Non-normal data, misspecified model				
Wald (Expected)	92.2	1.6	6.2	0.099
Wald (HW)	93.5	0.9	5.6	0.102
Monte Carlo (HW)	94.7	1.3	4.0	0.103
IJ1 percentile	94.3	1.4	4.3	0.102
HOIJ-2 percentile	94.6	2.0	3.4	0.102
Bootstrap percentile	94.9	1.9	3.2	0.104

Note. See Table 1.

Table 4: Simulation results for the structural disturbance variance ψ_{speed} at $N = 100$: coverage of nominal 95% intervals, tail balance, and median width, by condition and method (S = 1,000 data sets per cell, B = 1,000 shared bootstrap weight vectors).

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	91.8	0.4	7.8	0.516
Wald (HW)	92.2	0.6	7.2	0.525
Monte Carlo (HW)	91.9	0.7	7.4	0.526
IJ1 percentile	92.2	0.8	7.0	0.524
HOIJ-2 percentile	95.2	1.0	3.8	0.541
Bootstrap percentile	95.2	1.1	3.7	0.540
Normal data, misspecified model				
Wald (Expected)	92.8	0.4	6.8	0.510
Wald (HW)	92.3	1.0	6.7	0.521
Monte Carlo (HW)	92.3	1.1	6.6	0.514
IJ1 percentile	92.2	1.0	6.8	0.518
HOIJ-2 percentile	94.6	1.3	4.1	0.536
Bootstrap percentile	95.0	1.1	3.9	0.536
Non-normal data, correct model				
Wald (Expected)	88.7	1.4	9.9	0.496
Wald (HW)	89.2	0.2	10.6	0.513
Monte Carlo (HW)	89.0	0.3	10.7	0.509
IJ1 percentile	88.5	0.7	10.8	0.508
HOIJ-2 percentile	92.0	1.0	7.0	0.536
Bootstrap percentile	93.2	1.0	5.8	0.535
Non-normal data, misspecified model				
Wald (Expected)	90.1	1.8	8.1	0.487
Wald (HW)	90.2	0.3	9.5	0.493
Monte Carlo (HW)	90.1	0.3	9.6	0.491
IJ1 percentile	90.3	0.6	9.1	0.489
HOIJ-2 percentile	92.3	1.0	6.7	0.517
Bootstrap percentile	92.7	1.0	6.3	0.518

Note. See Table 1.

Table 5: Simulation results for the structural disturbance variance ψ_{speed} at $N = 200$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	92.6	1.0	6.4	0.372
Wald (HW)	93.3	0.8	5.9	0.373
Monte Carlo (HW)	93.0	1.0	6.0	0.371
IJ1 percentile	93.3	1.2	5.5	0.373
HOIJ-2 percentile	95.4	1.3	3.3	0.377
Bootstrap percentile	95.5	1.3	3.2	0.377
Normal data, misspecified model				
Wald (Expected)	92.5	0.8	6.7	0.371
Wald (HW)	92.3	0.8	6.9	0.375
Monte Carlo (HW)	92.0	1.1	6.9	0.373
IJ1 percentile	92.7	0.9	6.4	0.372
HOIJ-2 percentile	94.3	1.4	4.3	0.377
Bootstrap percentile	94.2	1.3	4.5	0.377
Non-normal data, correct model				
Wald (Expected)	90.7	0.8	8.5	0.368
Wald (HW)	89.7	0.2	10.1	0.371
Monte Carlo (HW)	90.6	0.2	9.2	0.372
IJ1 percentile	89.8	0.6	9.6	0.372
HOIJ-2 percentile	92.6	0.7	6.7	0.378
Bootstrap percentile	92.6	0.7	6.7	0.380
Non-normal data, misspecified model				
Wald (Expected)	92.1	1.4	6.5	0.361
Wald (HW)	91.6	0.3	8.1	0.366
Monte Carlo (HW)	91.5	0.3	8.2	0.364
IJ1 percentile	90.6	0.8	8.6	0.365
HOIJ-2 percentile	93.8	1.1	5.1	0.370
Bootstrap percentile	94.0	1.0	5.0	0.371

Note. See Table 1.

Table 6: Simulation results for the structural disturbance variance ψ_{speed} at $N = 500$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	95.2	1.1	3.7	0.241
Wald (HW)	95.3	1.0	3.7	0.243
Monte Carlo (HW)	95.1	1.2	3.7	0.241
IJ1 percentile	94.6	1.2	4.2	0.242
HOIJ-2 percentile	95.2	1.8	3.0	0.242
Bootstrap percentile	95.3	1.8	2.9	0.242
Normal data, misspecified model				
Wald (Expected)	93.7	2.1	4.2	0.243
Wald (HW)	93.5	2.1	4.4	0.241
Monte Carlo (HW)	93.2	2.2	4.6	0.238
IJ1 percentile	93.5	2.3	4.2	0.239
HOIJ-2 percentile	94.1	2.7	3.2	0.239
Bootstrap percentile	94.1	2.6	3.3	0.238
Non-normal data, correct model				
Wald (Expected)	92.7	1.6	5.7	0.240
Wald (HW)	92.9	0.8	6.3	0.248
Monte Carlo (HW)	92.1	1.0	6.9	0.246
IJ1 percentile	92.5	1.0	6.5	0.247
HOIJ-2 percentile	93.7	1.3	5.0	0.248
Bootstrap percentile	93.8	1.4	4.8	0.248
Non-normal data, misspecified model				
Wald (Expected)	93.6	1.8	4.6	0.238
Wald (HW)	94.6	0.8	4.6	0.241
Monte Carlo (HW)	94.5	0.9	4.6	0.242
IJ1 percentile	94.7	1.0	4.3	0.242
HOIJ-2 percentile	94.5	1.8	3.7	0.243
Bootstrap percentile	94.6	1.8	3.6	0.243

Note. See Table 1.

Table 7: Simulation results for the reliability coefficient ω_{speed} at $N = 100$: coverage of nominal 95% intervals, tail balance, and median width, by condition and method ($S = 1{,}000$ data sets per cell, $B = 1{,}000$ shared bootstrap weight vectors).

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	94.3	4.8	0.9	0.205
Wald (HW)	94.3	4.7	1.0	0.203
Monte Carlo (HW)	97.4	0.8	1.8	0.312
IJ1 percentile	97.1	1.0	1.9	0.308
HOIJ-2 percentile	98.1	1.8	0.1	0.266
Bootstrap percentile	96.6	3.0	0.4	0.209
Normal data, misspecified model				
Wald (Expected)	92.9	5.7	1.4	0.205
Wald (HW)	92.8	6.2	1.0	0.202
Monte Carlo (HW)	97.6	0.8	1.6	0.302
IJ1 percentile	97.3	1.0	1.7	0.291
HOIJ-2 percentile	97.9	2.0	0.1	0.261
Bootstrap percentile	95.1	4.2	0.7	0.209
Non-normal data, correct model				
Wald (Expected)	88.5	10.1	1.4	0.202
Wald (HW)	91.2	8.2	0.6	0.217
Monte Carlo (HW)	97.3	2.4	0.3	0.321
IJ1 percentile	97.0	2.3	0.7	0.304
HOIJ-2 percentile	95.2	4.6	0.2	0.267
Bootstrap percentile	92.9	6.8	0.3	0.221
Non-normal data, misspecified model				
Wald (Expected)	86.2	11.8	2.0	0.199
Wald (HW)	89.3	10.2	0.5	0.216
Monte Carlo (HW)	97.0	2.5	0.5	0.319
IJ1 percentile	96.7	2.6	0.7	0.296
HOIJ-2 percentile	94.5	5.5	0.0	0.263
Bootstrap percentile	91.5	8.2	0.3	0.220

Note. See Table 1.

Table 8: Simulation results for the reliability coefficient ω_{speed} at $N = 200$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	94.2	4.3	1.5	0.148
Wald (HW)	93.8	4.6	1.6	0.146
Monte Carlo (HW)	94.3	1.1	4.6	0.177
IJ1 percentile	94.0	1.4	4.6	0.174
HOIJ-2 percentile	97.1	2.2	0.7	0.161
Bootstrap percentile	94.8	2.7	2.5	0.146
Normal data, misspecified model				
Wald (Expected)	96.7	2.4	0.9	0.148
Wald (HW)	96.2	2.5	1.3	0.147
Monte Carlo (HW)	94.5	1.0	4.5	0.177
IJ1 percentile	94.8	1.0	4.2	0.175
HOIJ-2 percentile	97.9	1.5	0.6	0.162
Bootstrap percentile	95.7	2.2	2.1	0.148
Non-normal data, correct model				
Wald (Expected)	89.4	7.0	3.6	0.147
Wald (HW)	93.2	5.2	1.6	0.162
Monte Carlo (HW)	96.1	1.9	2.0	0.193
IJ1 percentile	94.9	2.3	2.8	0.188
HOIJ-2 percentile	94.9	4.3	0.8	0.174
Bootstrap percentile	93.5	5.0	1.5	0.162
Non-normal data, misspecified model				
Wald (Expected)	88.9	8.4	2.7	0.145
Wald (HW)	92.6	6.8	0.6	0.162
Monte Carlo (HW)	96.3	2.8	0.9	0.193
IJ1 percentile	95.5	2.8	1.7	0.188
HOIJ-2 percentile	94.3	5.3	0.4	0.173
Bootstrap percentile	93.4	6.0	0.6	0.161

Note. See Table 1.

Table 9: Simulation results for the reliability coefficient ω_{speed} at $N = 500$.

Method	Coverage	Miss left	Miss right	Median width
Normal data, correct model				
Wald (Expected)	93.6	4.5	1.9	0.093
Wald (HW)	93.9	4.3	1.8	0.093
Monte Carlo (HW)	94.0	1.8	4.2	0.100
IJ1 percentile	94.2	1.9	3.9	0.099
HOIJ-2 percentile	94.8	3.0	2.2	0.096
Bootstrap percentile	94.3	3.1	2.6	0.093
Normal data, misspecified model				
Wald (Expected)	95.3	3.0	1.7	0.093
Wald (HW)	95.2	3.2	1.6	0.093
Monte Carlo (HW)	95.7	1.2	3.1	0.099
IJ1 percentile	95.3	1.5	3.2	0.099
HOIJ-2 percentile	95.8	2.5	1.7	0.096
Bootstrap percentile	95.3	2.8	1.9	0.093
Non-normal data, correct model				
Wald (Expected)	89.8	6.4	3.8	0.093
Wald (HW)	94.4	4.0	1.6	0.107
Monte Carlo (HW)	95.4	2.5	2.1	0.114
IJ1 percentile	94.6	2.5	2.9	0.113
HOIJ-2 percentile	95.1	3.6	1.3	0.108
Bootstrap percentile	95.0	3.6	1.4	0.107
Non-normal data, misspecified model				
Wald (Expected)	89.1	7.5	3.4	0.093
Wald (HW)	94.3	4.7	1.0	0.106
Monte Carlo (HW)	95.7	2.9	1.4	0.113
IJ1 percentile	94.7	3.1	2.2	0.111
HOIJ-2 percentile	94.3	4.5	1.2	0.107
Bootstrap percentile	93.9	4.7	1.4	0.106

Note. See Table 1.